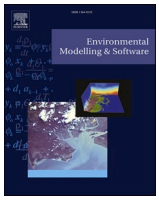
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Towards democratized flood risk management: An advanced AI assistant enabled by GPT-4 for enhanced interpretability and public engagement

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ABSTRACT

Traditional flood risk communication fails to bridge the gap between complex technical data and the needs of the public, hindering effective response. This research addresses this gap by developing and validating a novel AI-powered assistant that uses GPT-4 to democratize flood risk information. Our core methodology includes a Retrieval-Augmented Generation (RAG) framework that synthesizes real-time flood warnings, geospatial data, and social vulnerability indices into clear, conversational responses. To validate its effectiveness, we conducted a mixed-methods evaluation, including a comparison across different GPT models. Key quantitative findings reveal that the assistant achieved high performance scores in general flood knowledge (5/5) and handling flash flood alerts (4.3/5). Response times averaged a rapid 12 s for non-function-calling queries, though more complex data retrieval tasks averaged 36 s, highlighting areas for optimization. Our comparison identified GPT-4o as the optimal model for balancing accuracy with response time. The broader implications of this work demonstrate that large language models can serve as powerful tools to translate complex environmental data for non-experts, paving the way for more equitable, engaging, and effective public participation in disaster risk management.

1. Introduction

Effective communication is a cornerstone of flood risk management, important for at-risk populations to ensure resilience (Maskrey et al., 2016; Argent et al., 2016). While flood maps are a key tool intended to enhance public understanding and promote preparedness (Arrighi et al., 2019; Henstra et al., 2019), their utility as decision-support tools is often limited. A significant gap in public awareness persists because non-experts struggle to effectively utilize or understand flood maps (Leskens et al., 2014). This failure stems from several challenges, including the difficulty of balancing simplicity with necessary complexity (Hagemeier-Klose and Wagner, 2009), the public's struggle to interpret probabilistic information like flood return periods (O'Sullivan et al., 2012; Strathie et al., 2017), and a lack of direct flood experience, which hampers risk visualization, especially in dike-protected areas (Bradford et al., 2012; Botzen et al., 2009; Bubeck et al., 2012). Second, the traditional method of communicating flood risks through maps often follows a top-down strategy. This means that stakeholders are seen merely as receivers of information, with little to no involvement in creating the flood maps themselves. This approach can lead to flood maps that don't align with the actual needs of the users, since the content is not tailored to those it is meant to serve. This disconnect is a significant hurdle in communicating flood risks effectively (Meyer et al., 2012). A user-specific approach is required to improve the relevance and utility of flood maps. Third, Rollason et al. (2018) highlight that stakeholders are particularly interested in understanding how predicted flood levels might affect their properties specifically, and how these predictions compare to previous flooding events. This underscores the need for flood communication to be more interactive and responsive to user needs.

Risk communication is also a key factor that shapes how people perceive and respond to flood risks (Hagemeier-Klose and Wagner, 2009; Strathie et al., 2017). The resilience of a community post-flood is closely tied to human perception, which is itself influenced by the social context surrounding the flood event (Bodoque et al., 2016, 2019), because they link emotions, risk perceptions, and behavioral responses to floods. Understanding how people—be it residents, businesses, or government bodies—perceive and comprehend flood risks is vital to the decisions they make and the actions they anticipate.

In recent years, flood risk management shifts towards incorporating

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real-time forecasting, which has added further complexity to the field practice. To effectively merge real-time forecasting with current practices, there is a need for more rigorous and prompt coordination of emergency responses, as well as more dynamic engagement with the public. Traditional decision-making processes are based on complex numerical models that, while accurate after calibration and validation, can introduce delays that impede rapid decision-making. The complexity is amplified by the dynamic mixture of socio-cultural, institutional, and environmental factors that create hurdles in the clear communication of flood risks and hamper public participation. Furthermore, demographic shifts and the intensifying effects of climate change aggravate these challenges, underscoring the urgent demand for innovative technology solutions. Such advancements are necessary to strengthen the resilience of communities to the growing menace of flood-related disasters.

These issues become more challenging in the international background: flood risk communication is compounded in multicultural societies and regions with significant migrant populations, where one-way, single-language broadcasts fail to overcome linguistic and cultural barriers, diminishing trust and the capacity for self-protection (Tapsell et al., 2010). Furthermore, risk information is frequently fragmented across multiple agencies and presented in static formats that lack the personalization needed to motivate individual action (Forsyth, 2023). A useful tool is needed as a universal translator, both of complex scientific language into simple, actionable advice and of official warnings into numerous languages to empower communities worldwide.

The invention of Generative Artificial Intelligence, particularly Large Language Models (LLMs), introduces a revolutionary opportunity with the potential to transform flood risk management. These models have the remarkable ability to generate text that mirrors human writing, producing everything from poetry (Köbis and Mossink, 2021) to political statements (Bullock and Luengo-Oroz, 2019) and scholarly papers (Hu, 2023). The application of Generative AI extends beyond text, as it has been used to create visual art (Gillotte, 2019), synthetic human faces (Whittaker et al., 2020), and even detailed magnetograms of the Sun (Liu et al., 2021), demonstrating its extensive range of capabilities across various disciplines. LLMs hold promise as a versatile tool that could bridge the gap between the complex integration of expert numerical modeling and the practical needs of non-expert stakeholders. This could enable more effective communication and decision-making processes in flood risk management by providing an accessible platform for public engagement and understanding. The application of LLM has the potential to enhance the experience of flood map users, provide an interactive platform for two-way communication, improve the efficiency and effectiveness of integrating real-time flood warning, and, more importantly, reshape the perception of flood risk and preparedness.

It is worth noting that AI has been employed in flood research for several years (Wang, 2021). Early efforts demonstrated how computer vision could leverage social media imagery to pinpoint flood locations, effectively transforming smartphones into real-time data collection tools (e.g., Wang et al., 2018). While the reliability of this approach has faced challenges, particularly after Twitter (now X) removed precise location-sharing (Hu and Wang, 2020), AI has become integral to flood modeling and mapping (Wang and Ding, 2023). Subsequent applications expanded into predictive modeling (Chang et al., 2020), risk assessment (Pham et al., 2021), and mitigation planning (Ye et al., 2021). However, these efforts primarily focused on providing data *to* decision-makers, with a persistent gap in creating two-way, interactive communication, as older systems could not adaptively respond to unstructured questions. The recent rise of Large Language Models (LLMs) offers a promising solution to this long-standing challenge. This is a rapidly advancing field, evidenced by several overview papers and new studies published during the peer-review process of this manuscript alone (e.g., Lei et al., 2025; David et al., 2024; Xie et al., 2025a, 2025b; Zajac et al., 2025; Li et al., 2025).

This study explores the untapped potential of Large Language Models (LLMs)—specifically, the Generative Pre-trained Transformer 4 (GPT-4)—to address the aforementioned challenges in flood risk communication. While GPT-4's effectiveness is well-documented across sectors like education and medicine, its adaptation to the complex scenarios of flood risk management remains a novel area of exploration (Liu et al., 2023). Our primary contribution is the creation of a specialized AI Assistant designed to test the feasibility of using LLMs to: 1) generate easy-to-understand flood maps, 2) involve users in the flood map generation process, and 3) translate specialized flood risk data for public communication. To achieve this, we leverage the distinctive "function calling" feature of GPT-4, which equips our Flood AI Assistant with the ability to access external GIS APIs and query social vulnerability datasets. This paper is structured as follows: Section 2 introduces our methodology, Section 3 presents the research results, Section 4 details various applications, and Section 5 discusses limitations and future research directions.

2. Method

2.1. AI assistant framework

To develop such an AI Assistant, we designed four specialized functions, namely:

(1) *get_flood_map:* a function that provides valuable flood risk information by generating both a static flood map and an interactive map using standard latitude and longitude coordinates. The static map is color-coded with FEMA Flood Zone designations, enabling users to assess the risk associated with different regions. In case of any issues, an error message will be returned, ensuring transparency about potential retrieval errors for the static map. The interactive map allows users to explore flood risk dynamically.
(2) *get_flood_data:* a function that returns information related to flood risk and property details based on provided addresses. The response includes details such as flood hazard area, FEMA FIRM (Flood Insurance Rate Map) panel information, elevation, political and community information, census data, and geocoding details like latitude and longitude.
(3) *get_svi_stats_and_tracts:* a function retrieves Social Vulnerability Index (SVI) statistics and information about census tracts based on user-specified conditions, including state abbreviation, county, SVI theme, a comparison operator, and a threshold value. It answers questions related to the vulnerability of specific areas by providing statistics such as the total number of vulnerable areas, maximum SVI score, minimum SVI score, and average SVI score, along with details (including boundary coordinates) for each relevant census tract in JSON format. With the interface, the AI Assistant can also display an interactive map with the census tracts of interest for the user to explore. In the prototype we used the 2020 US SVI data (*CDC/ATSDR SVI data and documentation download*, 2022).
(4) *get_flash_flood_warnings:* a function retrieves active flood alerts from the National Weather Service's API based on the specified location. It returns details about the flood alerts, including the event type, description, timestamp sent, timestamp expires, and area description.

With these functions, we design the framework of the AI Assistant This diagram (Fig. 1) provides an overview of how the GPT-4 API processes user prompts with the aid of specialized function calls within the Flood AI Assistant:

1) **User Prompt Submission:** The chat application sends user's queries, along with a list of available functions, to the GPT-4 API.

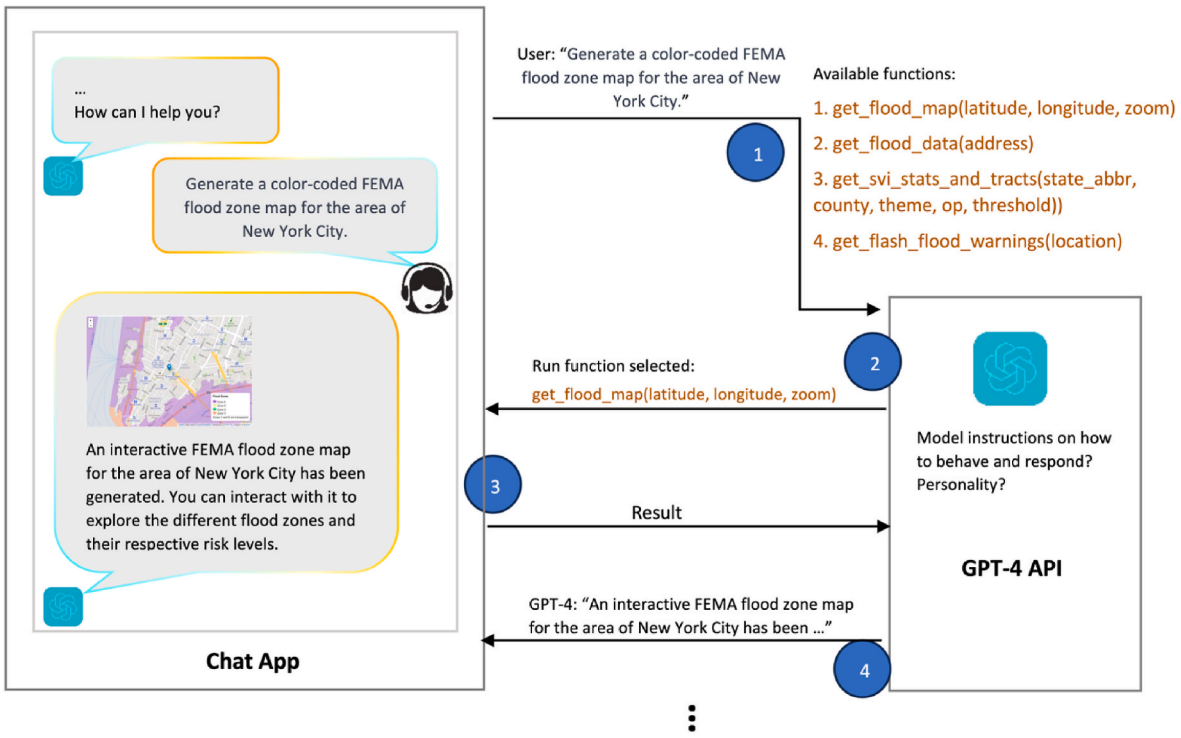


Fig. 1. Diagram illustrating GPT-4 API interaction with function calling in the flood AI assistant prototype.

2) **Model Response and Decision:** The GPT-4 model evaluates the query and either generates a text response directly or identifies the need for a specific function call.
3) **Function Execution:** If a function call is required, the AI Assistant executes the requested function, harnessing the relevant data or processing the necessary computation, and then returns the results to the GPT-4 API.
4) **Response Generation and Iteration:** Utilizing the data from the executed function, the model crafts a coherent text response. In certain cases, the API may initiate a subsequent function call for additional information or clarification.

A detailed flowchart about the procedure of the AI assistant is shown in Fig. 2.

2.2. Evaluation methodology

Our study evaluated the performance of our flood-related AI prototype using a structured approach. The evaluation focused on six key criteria, categorized under three primary factors: Relevance, Error Resilience, and Context Understanding.

The performance criteria include the following aspects:

1) Relevance: "***Accuracy***" assesses if the AI's responses are factually correct, relevant to the query, and free from misinformation. "***Completeness***" determines whether the response fully addresses all aspects of the query. **2) Error Resilience:** "***Error Handling***" evaluates the AI's capability to manage errors, uncertainties, or incomplete data, including its ability to recognize and communicate these issues. "***Informative Responses***" rates the AI's effectiveness in providing useful information, especially when data is unavailable, or the query is ambiguous. **3) Context Understanding:** "***Appropriateness***" checks if the response is suitable and relevant to the query's context. "***Adaptability***" measures the AI's flexibility in modifying responses based on the specifics of the query or evolving conversation.

These criteria were applied to assess the prototype's responses to a set of predefined prompts. The prompt categories include Get_Flood_Map Functions, Get_Flood_Data Functions, Get_SVI_Stats_And_Tracts Functions, Get_Flash_Flood_Warnings Functions, and General Understanding of Flood-Related Queries. Each category was designed to test the prototype's specialty functions and its general comprehension of flood-related scenarios, leveraging the GPT-4 model's pre-trained knowledge base.

We conducted two iterations of testing with six prompts in each of the five prompt categories. The original prompts were designed to test the performance of both ambiguous and more specific queries for each prompt category. The initial iteration was used to identify prompts requiring refinement to maximize the prototype's use of its specialty functions. Iteration 2 was used to adjust the initial prompt and evaluate the performance. Each response was scored on a scale of 1–5 for all criteria, with the following qualitative interpretations: 1/5 (Poor): Significantly unsatisfactory in meeting the criteria. The response may be largely irrelevant, inaccurate, or not useful. 2/5 (Fair): Partially meets the criteria but has notable deficiencies. The response might be somewhat relevant or useful but lacks key elements or clarity. 3/5 (Good): Adequately meets the criteria. The response is relevant and useful, but there is room for improvement in certain aspects. 4/5 (Very Good): Strongly meets the criteria with only minor shortcomings. The response is highly relevant, accurate, and largely fulfills the user's needs. 5/5 (Excellent): Exceptionally meets the criteria. The response is comprehensive, highly accurate, thoroughly informative, and fully addresses the user's query.

2.3. Computational configuration

Our methodology centers on leveraging the inherent capabilities of a pre-trained Large Language Model (LLM), and it is important to state explicitly that the model was **not retrained** or fine-tuned on a new dataset. Instead, the AI Assistant's specialized behavior is guided entirely through "in-context learning," where its performance is shaped by the information provided to it at the moment of the query. This is achieved through systematic **prompt engineering**, where a foundational set of instructions defines the AI's role and constraints. For

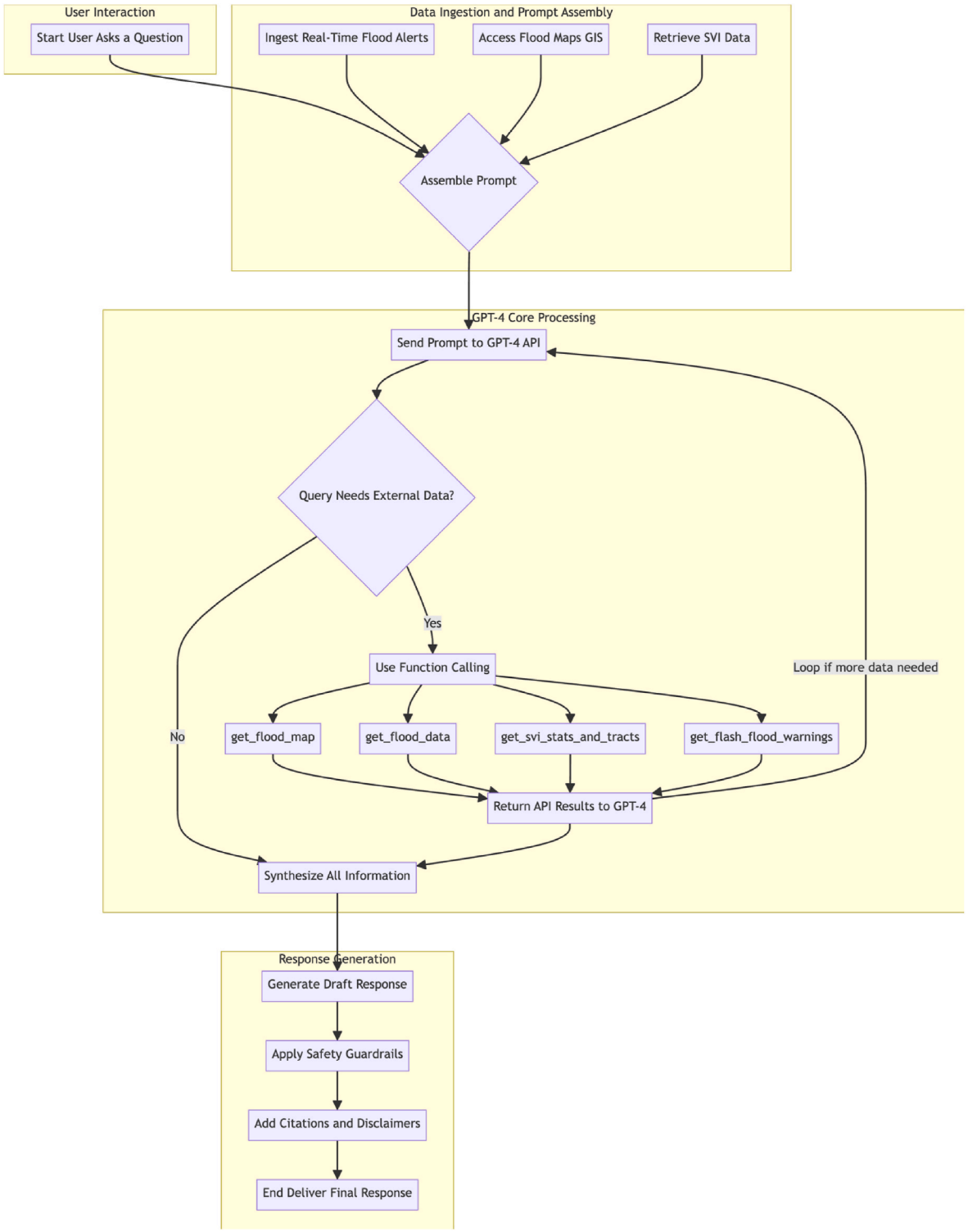


Fig. 2. Flowchart of the procedure used in the AI assistant focusing on the flow of data processing.

example, a core instruction in the system prompt is: *"You are an AI assistant for flood risk communication. Your goal is to help users understand flood maps and alerts. Always translate technical terms into simple language and cite your data sources."* This initial guidance is complemented by providing real-time, relevant data—such as a specific flood alert or GIS data—directly within the user's query, which grounds the model's answers in factual, current information rather than its generalized pre-trained knowledge.

The effectiveness of this approach hinges on three core capabilities of the GPT-4 architecture. First, its **advanced natural language understanding** allows it to interpret a user's intent from informal or ambiguous questions, making it highly accessible to the public. Second, its capacity for **information synthesis** enables it to take multiple, disparate pieces of information provided in the prompt—such as a user's location, a technical flood alert, and social vulnerability data—and weave them into a single, coherent, and contextually relevant paragraph. Finally, the model excels at **in-context reasoning**, performing logical steps based solely on the data it is given. It can deduce if a user's

address falls within a flood zone defined in the provided data and explain that conclusion in simple terms.

For our evaluation, we utilized the "gpt-4-1106-preview" model with default temperature settings to balance creativity and factual relevance. It is crucial to acknowledge the model's operational constraints: its training data is limited to information up to April 2023, and it has a maximum output of 4096 tokens. This token limit necessitates careful management of conversation history to avoid losing context. Understanding these technical parameters is essential for interpreting the assistant's performance and guiding future development toward a more robust and efficient tool for flood risk management.

The evaluation of the model involved designing *one-shot* prompts to test the specialty functions under specific, ambiguous, or vague queries. The numbering system identifies the prompt category, while single, double, or triple letters indicate the stages of prompt refinement. For example, "3B" is the second iteration of prompt B in Category 3, and "3BB" is a further refinement of "3B." Detailed information on these prompts, their responses, and individual criteria scores are available in Appendix A, Tables A1 and A2. It is worth noting our analysis focuses on Iteration 2, because Iteration 1 is preliminarily experimental for prompt refinement and consistency examination.

With original prompts, Categories 2, 4, and 5 did not require prompt refining to thoroughly study the prototype's performance with these categories. Category 2 demonstrated the model's competence in responding to queries for flood data for specific addresses and its ability to request additional information. Category 4, focusing on flood alerts, showcased the model's performance with original prompts, so further refinement is unnecessary. For general flood-related queries, i.e.,

Case Study 1: New Orleans Flood Alert and Vulnerability Assessment.

What are the current flood alerts for the New Orleans area?

Currently, there are no flash flood warnings issued for the New Orleans area in Louisiana. Please note that flood situations can change rapidly, so it's always a good idea to stay informed by checking for updates regularly, especially during heavy rain events or severe weather conditions. If you need further assistance or more detailed information, feel free to ask.

Can you identify the neighborhoods in New Orleans that have historically faced the most severe flooding?

To identify the neighborhoods in New Orleans that have historically faced severe flooding, we can consider areas within Special Flood Hazard Areas (SFHAs), as well as those affected by hurricanes and severe storms in the past. Flood maps, historical data, and local knowledge can provide us with this information.

I can provide you with a general idea of the neighborhoods based on historical events, notably from Hurricane Katrina, which hit New Orleans in 2005. Some of the neighborhoods that were most severely affected by flooding at that time include:

- The Lower Ninth Ward: This neighborhood suffered extensive damage due to the levee failures, with many homes completely destroyed by the floodwaters.

- Lakeview: Located near Lake Pontchartrain, Lakeview also experienced severe flooding from levee breaches.

- New Orleans East: This area was inundated due to its proximity to the Industrial Canal and the Gulf Intracoastal Waterway.

- Gentilly: This neighborhood, located on the edge of the city, is low-lying and was significantly impacted by flooding.

- St. Bernard Parish: Although not a neighborhood of New Orleans, St. Bernard Parish, adjacent to the city, was almost entirely flooded.

Current flood maps can also show areas that are at risk of flooding based on latest data. I can fetch a flood map for New Orleans to show the designated flood zones, which will give us a visual representation of the areas at risk. Would you like to proceed with this?

For the neighborhoods you mentioned, can you provide SVI statistics for areas with an SVI score higher than 0.9?

Case Study 2: Baton Rouge Construction Site Flood Risk Analysis.

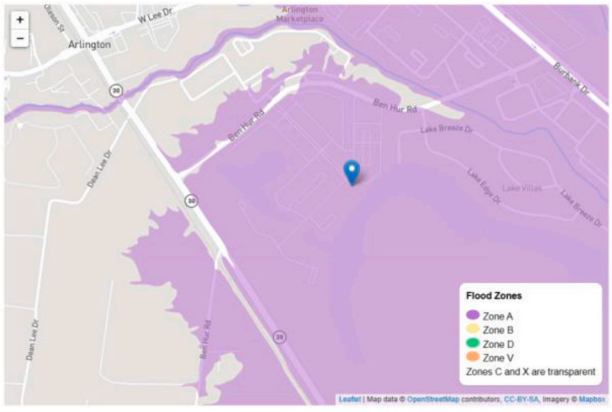
Is there a flood risk for the proposed construction site at 777 Ben Hur Rd Baton Rouge, LA 70820 in Baton Rouge?

The construction site at 777 Ben Hur Rd, Baton Rouge, LA 70820 is located within a flood zone designated as "AE." This classification indicates that the area is at a high risk for flooding, as it is within the Special Flood Hazard Area (SFHA), where there is a 1% annual chance of flood hazard. The designation "T" for SFHA_TF also affirms that it is in a regulatory floodplain.

These zones are areas subject to inundation by the base flood, which is the flood having a 1% chance of being equaled or exceeded in any given year. It's also known as the 100-year flood. Construction in this area will likely need to comply with local and federal floodplain management regulations, such as raising buildings above the base flood elevation.

It's important for anyone planning construction in this area to consult with local building officials, FEMA floodplain maps, and potentially an engineer specializing in hydraulics or hydrology for specific construction requirements and mitigation strategies.

Can you provide a detailed flood zone map for this area to help us understand the specific flood risks?

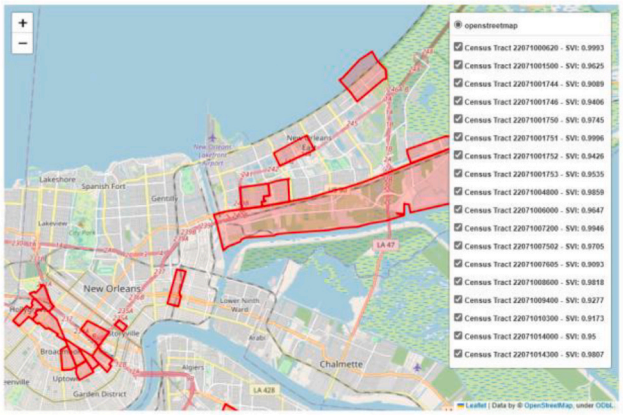


I apologize for the inconvenience; there seems to be an internal error while retrieving the static flood map. However, you can view the interactive flood map for the location at 777 Ben Hur Rd, Baton Rouge, LA 70820 by following the link provided below. This map will show the flood zones and help you understand the specific flood risks for the area.

[View Interactive Flood Map for 777 Ben Hur Rd, Baton Rouge, LA 70820](sandbox:/static/interactive_flood_map.html)

To get a detailed understanding of the flood risks in this area, you can also refer to FEMA's official flood maps, or contact local authorities for the most accurate and up-to-date information on flood zones and

Fig. 3. Example conversation records of the case studies with details of prompts and sample AI responses.



In the Orleans County area of New Orleans, there are several census tracts with SVI scores higher than 0.9, indicating a high level of social vulnerability. Here are the areas along with their respective SVI scores:

1. Census Tract 22071000620 - SVI Score: 0.9993
2. Census Tract 22071001500 - SVI Score: 0.9625
3. Census Tract 22071001744 - SVI Score: 0.9089
4. Census Tract 22071001746 - SVI Score: 0.9406
5. Census Tract 22071001750 - SVI Score: 0.9745
6. Census Tract 22071001751 - SVI Score: 0.9996
7. Census Tract 22071001752 - SVI Score: 0.9426
8. Census Tract 22071001753 - SVI Score: 0.9535
9. Census Tract 22071004800 - SVI Score: 0.9859
10. Census Tract 22071006000 - SVI Score: 0.9647
11. Census Tract 22071007200 - SVI Score: 0.9946
12. Census Tract 22071007502 - SVI Score: 0.9705
13. Census Tract 22071007605 - SVI Score: 0.9093
14. Census Tract 22071008600 - SVI Score: 0.9818
15. Census Tract 22071009400 - SVI Score: 0.9277
16. Census Tract 22071010300 - SVI Score: 0.9173
17. Census Tract 22071014000 - SVI Score: 0.95
18. Census Tract 22071014300 - SVI Score: 0.9807

These tracts represent areas within New Orleans that are more likely to need support during emergencies, including flooding, due to factors such as socioeconomic status, household composition, race, ethnicity, and housing/transportation characteristics.

Please find the map below showing the highlighted census tracts with high SVI scores:

[Interactive SVI Map with Census Tracts Highlighted](sandbox:/static/svi_tracts_and_stats_map.html)

Keep in mind that the SVI scores reflect more than just flood risk—they encompass various aspects of vulnerability that can heighten the impact of disasters.

requirements for building in flood-prone areas.

Based on the flood risk and zone information, what mitigation measures should we consider for this construction project?

Given that the proposed construction site at 777 Ben Hur Rd, Baton Rouge, LA 70820 is within a flood zone designated as "AE," it is important to consider the following mitigation measures for the construction project:

1. **Elevation**: Elevate the building above the base flood elevation (BFE) as outlined by FEMA for zone AE. This might involve the use of stilts, piers, or fill to raise the structure's lowest floor.

2. **Foundation**: Design a foundation that can resist flood forces, such as deep foundations that extend to stable soils below the zone of influence of floodwaters, or pile foundations that can resist lateral forces.

3. **Floodproofing**: Make the building watertight below the BFE through methods like sealing walls with waterproof coatings, using flood-resistant materials, installing backflow valves to prevent sewage system backups, and using sump pumps.

4. **Site Planning**: Orient and position the building to minimize exposure to flowing water. Preserve natural features that absorb floodwaters, and avoid construction in areas that would be cut off by flooded access routes.

5. **Stormwater Management**: Implement green infrastructure, like rain gardens and permeable pavement, to manage runoff and reduce the potential for flooding on and around the site.

6. **Flood Barriers**: Consider temporary flood barriers or levees that can be deployed when flooding is imminent to protect the structure and site.

7. **Utilities and Services**: Elevate and secure mechanical, electrical, and plumbing systems above the BFE, and use flood-resistant materials for components that cannot be elevated.

8. **Emergency Action Plan**: Develop an emergency response plan for occupants that outlines evacuation routes, procedures, and safety measures in case of a flood.

9. **Landscaping**: Utilize landscaping to help divert floodwaters away from buildings and reduce erosion.

10. **Insurance**: Ensure that the building is adequately insured under the National Flood Insurance Program (NFIP), especially since buildings within the AE zone are usually required to have flood insurance if they are financed by a federally insured or regulated lender.

11. **Regulatory Compliance**: Comply with local floodplain management ordinances and zoning codes, which may have stricter requirements than the federal regulations.

Incorporating these measures can help protect the building and its future occupants from flood risks. Consulting with a civil engineer specializing in hydraulics and flood prevention, as well as local authorities, will be crucial for developing an effective flood mitigation strategy that meets all local, state, and federal requirements.

Fig. 3. (*continued*).

Category 5, the model exhibited a robust understanding and handling of a diverse range of topics.

2.4. Comparison with other GPT models

The latest GPT models were used to assess the accuracy and the performance of the output for a comparative study against the main model used in the study (GPT-4-1106-preview). The response evaluation is the same as using the metrics above for two GPT models: GPT-4o ("o" for "omni") is the most advanced multimodal model, that is faster and cheaper than GPT-4 Turbo with stronger vision capabilities — it generates text two times faster and is 50 % lower in token costs. GPT-4o-mini a cost-efficient small model, which is more intelligent and cost-efficient than GPT-3.5 Turbo with vision capabilities. GPT-4o-mini

was recommended to replace GPT-3.5-turbo.

3. Results

3.1. Case study

Here we present two illustrative examples (Fig. 3) to showcase the inputs and outputs for the system. These cases are aimed to give a concrete and specific overview of the system application to facilitate the understanding of the evaluation results.

Case Study 1 was designed to demonstrate the top-bottom scenario of flood information inquiry. Specifically, the questions started with an inquiry of the general region and then asked for information about specific addresses. This case study focuses on New Orleans, which has a profound and complex risk profile. Geographically, much of the city sits below sea level in a bowl-like depression and is experiencing significant land subsidence. Hydrologically, it is positioned between the Mississippi River and Lake Pontchartrain, making it vulnerable to riverine flooding, intense rainfall, and hurricane-induced storm surges. This acute physical exposure is compounded by significant social vulnerability, as historical events like Hurricane Katrina have disproportionately impacted low-income and minority communities located in the most at-risk, low-lying areas. This intersection of extreme physical and social risk makes New Orleans a critical environment for testing advanced risk communication tools designed to reach diverse and vulnerable populations.

The case study demonstrates the AI Assistant's ability to conduct a multi-layered risk assessment that integrates real-time alerts, historical context, and geospatial datasets. The interaction began with a query for current flood alerts, then pivoted to identifying historically impacted neighborhoods like the Lower Ninth Ward. The system's core capability was showcased when it leveraged its "function calling" feature to programmatically query an external Social Vulnerability Index (SVI) database. It successfully identified and listed 18 specific census tracts with SVI scores higher than 0.9 and generated a final interactive map visualizing these highly vulnerable areas. This workflow highlights the system's capacity to move beyond simple inquiries to perform complex data retrieval, synthesis, and visualization, thereby providing a more holistic and actionable risk assessment.

Case Study 2 was designed to test the bottom-top scenario in Baton Rouge, which serves as a critical case study for assessing flood risk in inland, riverine communities experiencing rapid suburban development. While parts of the city are situated on natural bluffs along the Mississippi River, much of the surrounding East Baton Rouge Parish is low-lying and susceptible to fluvial flooding from the Amite and Comite Rivers and their tributaries. This risk was tragically realized during the catastrophic rainfall event of 2016, which inundated tens of thousands of homes, many of which were not in previously designated high-risk flood zones. The region's hydrology is complex, influenced by both river backwater and intense localized rainfall overwhelming drainage systems. This physical vulnerability is exacerbated by ongoing land-use changes, where development into flood-prone areas increases impervious surfaces and alters natural drainage patterns, creating a dynamic and escalating risk profile that demands site-specific analysis for new construction.

This case study showcases the AI Assistant's utility in providing detailed, site-specific flood risk analysis for engineering and planning purposes. The user provided an address for a proposed construction site, and the assistant immediately identified its location within a high-risk "AE" flood zone, or Special Flood Hazard Area (SFHA). It correctly interpreted this designation as having a 1 % annual chance of flooding and noted that construction would be subject to federal and local regulations. While the system encountered a temporary internal error preventing the generation of a static map, it successfully provided a link to an interactive version. Critically, the assistant then synthesized the risk information into a comprehensive list of actionable mitigation measures. These included specific recommendations such as elevating the structure above the Base Flood Elevation (BFE), designing flood-resistant foundations, implementing green infrastructure for stormwater management, and ensuring compliance with the National Flood Insurance Program (NFIP). This demonstrates the tool's ability to translate raw geospatial data into practical, expert-level guidance for resilience planning.

These scenarios exemplify the AI Assistant's remarkable capacity to synthesize diverse and unstructured flood-related data into coherent, actionable insights. Whether it is merging real-time flood alerts with historical data and social vulnerability indices or translating flood zone categorizations into practical construction advice, the Assistant proves invaluable. Its ability to guide users from problem identification through to solution formulation is a testament to its potential as a transformative tool in flood risk management. By offering a nuanced understanding that combines technical data with socio-economic considerations, the AI Assistant empowers users to make well-informed, context-sensitive decisions, paving the way for more resilient and equitable responses to flood risks.

3.2. Performance evaluation

In this section, we computed the average scores across criteria per prompt categories to evaluate the prototype's performance in each predefined function and general flood-related knowledge (Table 1). For Categories 1 and 3, both the original and refined prompt scores were included to reflect the shortcomings of these functionalities. Error Resilience emerged as the lowest-scoring criteria across the predefined functions, primarily due to inadequate informative responses in error situations or when handling vague queries. This factor also influenced the adaptability of the Assistant, impacting its completeness score. Notably, the categories of general flood understanding, and flash flood alerts exhibited the highest performance, achieving overall scores of 5 and 4.3, respectively. In contrast, the flood map and SVI stats functions demonstrated lower performance, both achieving an overall score of 4. These findings suggest that enhancing function design for clarity, ease of use, and efficient token usage will be crucial in future development.

3.3. Response time analysis

Detailed response times for each prompt, as observed during iteration 2, can be referenced in the "Observations" column of Table A2 in Appendix A. The information about the time consumption in each inquiry provides another dimension to view of the response dynamics across different categories and reflect the computational resource demands.

An insightful pattern was revealed in the AI system's performance. When a function call was involved, the average response time was recorded at 36 s. The prompt category for flash flood alerts (Category 4) demonstrated the fastest response within this group, taking only 5 s. In contrast, the longest response time was observed in the SVI stats category (Category 3), which took up to 198 s by average, especially in instances where the model attempted but failed to retrieve data. Conversely, prompts that did not invoke a function had a significantly lower average response time of 12 s. The quickest responses in this group, typically around 3 s, were often observed when the model requested additional details from the user. The slowest response in this category, at 37 s, was noted in the general flood-related knowledge category (Category 5), where the responses were generally more comprehensive and detailed. A deeper dive into the category-specific response times reveals that Category 3 (SVI stats) had the longest average response time for function calls, averaging 85 s, with a range from 57 s to an extreme case of 198 s. This discrepancy highlights that functions retrieving or searching large datasets can require more processing time, underscoring the need for efficiency in function design.

In terms of the overall speed of the AI system, the average response times can be considered satisfactorily fast, especially given the

Table 1
Average scores across all six criteria and mean overall performance grouped by prompts category.

Prompt Category	Relevance		Error Resilience		Context Understanding		Overall Performance
	Accuracy	Completeness	Error Handling	Informative Responses	Appropriateness	Adaptability	
1	2.33	1.66	1.83	1.83	2	1.83	1.91
	1.99		1.83		1.91		
2	4.5	3.66	4.16	3.66	4	3.5	3.91
	4.08		3.91		3.75		
3	4.83	4.33	3.33	4.83	3.83	4.33	4.24
	4.58		4.08		4.08		
4	4.5	4.16	4.16	4.5	4	3.83	4.19
	4.33		4.33		3.91		
5	5	5	4	5	5	4	4.66
	5		4.5		4.5		

complexity of the tasks performed. The faster responses for simpler queries or when additional information is requested indicate the model's efficiency in handling straightforward tasks. However, the longer response times for more complex data retrieval tasks, like in the SVI stats category, suggest areas for optimization. Such response times, while still within reasonable bounds, could be improved with more efficient data handling or system enhancements.

It is important to contextualize these findings by acknowledging that response times can be influenced by external factors, such as server response time, internet speed, and GPT versions, which could account for some of the variability observed. These findings not only contribute to our understanding of the AI system's operational efficiency but also provide valuable insights for future improvements, particularly in optimizing the system for more rapid data processing without compromising the accuracy or comprehensiveness of the responses.

3.4. Comparison among GPT models

The average performance score and response time were used to compare the models in Fig. 4a. GPT-4o showed the highest accuracy with a moderate response time. In contrast, GPT-4-1106-preview had the lowest accuracy and the slowest response time, while GPT-4o-mini was the fastest but with reduced accuracy, as expected. Fig. 4b provides a more detailed comparison of response times, confirming that GPT-4o-mini consistently performed the fastest. These results suggest that GPT-4o is ideal when accuracy is the priority, while GPT-4o-mini excels in scenarios where quick responses are more important. On the other hand, GPT-4-1106-preview, as a preliminary model, requires further tuning to match the performance of the other models.

4. Discussion

Our research, utilizing GPT-4 models for flood risk management, has unveiled the significant potential of AI Assistants in bridging the gap between intricate flood data and practical, accessible information. Despite the challenges posed by the model's token limitations and the inherent complexity of flood-related queries, our prototype AI Assistant demonstrated considerable proficiency in interpreting and responding to a diverse range of flood-related questions. This is evident in the successful handling of specific and ambiguous queries, as well as its ability to relay complex data in an understandable format.

The application of AI in flood research is not new; however, its role has historically been different. Previous studies have effectively used AI for data mining from unstructured sources like social media to identify flood locations (Wang, 2021; Wang et al., 2018) or for advanced predictive modeling and risk assessment (Chang et al., 2020; Pham et al., 2021). While valuable, these applications primarily focus on providing data to decision-makers in a one-way flow. A persistent gap has remained in creating systems that can engage in a responsive, two-way dialogue, largely because older modeling systems could not adapt to unstructured, conversational questions.

The primary novelty of our work lies in leveraging the generative and reasoning capabilities of an LLM to create this interactive cycle. Unlike previous systems, our AI Assistant does not just present data; it interprets, synthesizes, and translates it in response to specific user queries. This marks a shift from AI as a data-mining tool to AI as a communication and interpretation partner. The global relevance of this approach is significant. Challenges in translating technical risk information for diverse, non-expert populations are universal (O'Sullivan et al., 2012). Our framework provides a scalable solution that can be adapted to different languages and contexts, offering a new paradigm for democratizing scientific information and enhancing public engagement in disaster preparedness worldwide.

To be viable beyond a research prototype, several factors should be further considered. First, the AI assistant's architecture is designed for high feasibility, as it leverages existing, mature technologies. It relies on

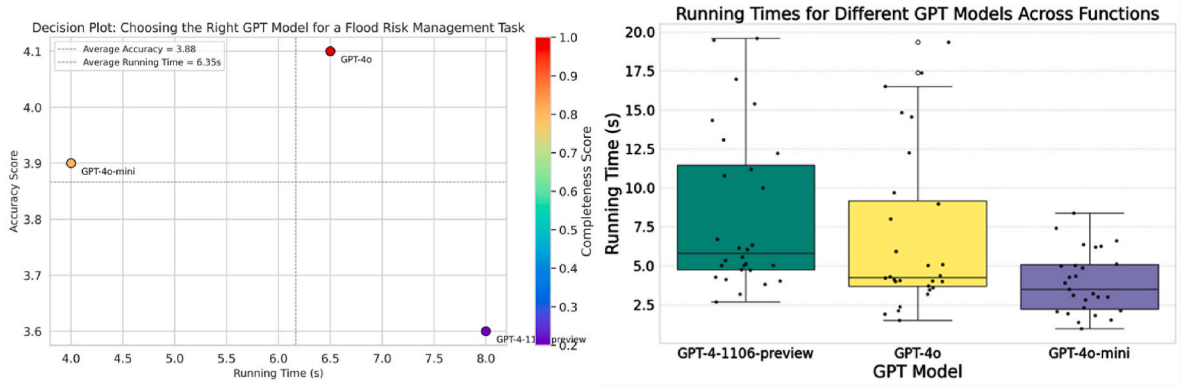


Fig. 4. Performance comparison among different models of GPT showing GPT-4o has the highest accuracy while GPT-4o is the fastest model.

API calls to the GPT-4 model and public data streams, avoiding the need for bespoke model development and focusing instead on the novel integration of proven components. Implementation for a municipality would primarily involve connecting the system to local, authoritative data feeds. Second, the system presents a highly cost-effective model for enhancing public communication. The primary operational costs—API usage and cloud hosting—are proportional to usage. Even during a high-demand flood event, the cost of automated, instantaneous responses is a fraction of the expense required to staff 24/7 call centers or dedicate public information officers to answer repetitive inquiries. The economic benefit lies in the potential for reduced property damage and human costs through more effective public preparedness. Third, the framework is inherently scalable. Its cloud-native design allows computational resources to expand automatically to meet high demand during emergencies. Furthermore, the system's modularity and the multilingual capabilities of the underlying LLM allow it to be readily adapted to new geographical regions and linguistic contexts with minimal modification.

In addition to the technical factors, deploying an AI system for high-stakes risk communication requires a robust ethical framework. First, we acknowledge that foundational datasets, such as the Social Vulnerability Index (SVI), can carry historical and social biases. Our framework does not resolve these underlying data issues but aims to mitigate their impact through transparency. Second, the assistant is designed to be a faithful interpreter of official data, not an originator of new facts. To ensure transparency and combat potential bias, our system employs several strategies: mandating source attributions for all data presented, clearly stating the currency of data layers (e.g., "using the 2020 SVI dataset"), and providing direct links back to the official sources whenever possible. Third, we define the AI Assistant as an advanced communication tool. As such, accountability for the information's content and the consequences of public actions rests not with the tool itself, but with the public agency that deploys it. This is analogous to the accountability for information published on an official government website or sent via an emergency alert system.

The detailed evaluation of the AI Assistant, although conducted with limited trials, revealed key strengths and areas for improvement. Notably, the Assistant's performance in general flood knowledge queries indicated the high value of using predefined functions, underscoring the need for clearer function instructions and design enhancements. The study also highlighted the Assistant's struggle with large data queries and its limited adaptability in response to errors and vague queries. These insights are invaluable in guiding future iterations of the system.

We would also like to comment on the error handling capability of the AI assistant. In Category 1 (Get_Flood_Map Function), the function's shortage to retrieve static maps due to server issues provided a valuable opportunity to evaluate the model's error handling. We observed that while the model effectively communicated errors, it often missed opportunities to enhance user understanding, such as elaborating on color-coded flood map areas. Fig. 5 shows mainly improved error resilience through prompt refinements, such as directly requesting specific locations or rephrasing prompts. However, Relevance and Context Understanding were also generally improved. The main improvement was seen in prompt D's refinement, while E only improved Error Resilience. Prompt F slightly improved across all three main criteria.

For Category 3 (Get_SVI_Stats_And_Tracts Function), the model faced challenges handling large data queries, often leading to the overlooks of crucial context or function response due to exceeding the maximum token output limit. To address this, we refined prompts to narrow down requested areas and social vulnerability scores. However, significant challenges remained due to the function's tendency to retrieve extensive data, which limits its practical utility. In real-world applications, alternative display methods for census tract data could be implemented to circumvent token limitations. Fig. 6 illustrates that refining prompts to specify smaller score ranges or directly ask for a model-suggested area of interest notably improved the model's performance. The first refinement of Prompt D initially performed worse, but a second refinement

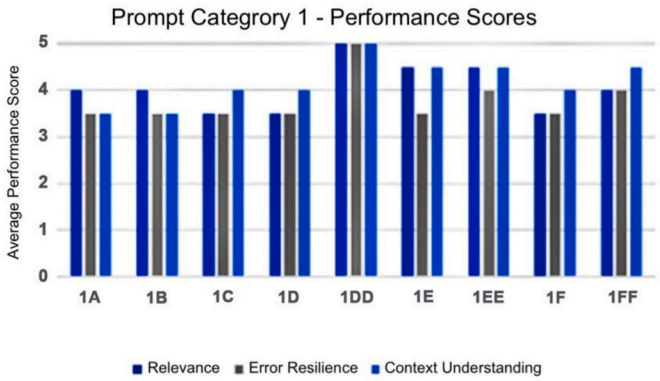


Fig. 5. This graph illustrates the average performance scores across three main evaluation criteria: Relevance, Error Resilience, and Context Understanding. It specifically compares the performance improvement for prompts D, E, and F following their refinement. The data highlights how targeted adjustments to the prompts led to enhanced overall performance of the AI system, demonstrating its adaptability and responsiveness to more refined user inputs.

considerably improved the performance, outperforming the original prompt. Refinements of prompts B, E, and F resulted in perfect scores in all 3 criteria. These observations highlight the importance of clear and informative function instructions for the model, and the need for better-designed functions overall.

Also, the prototype's performance was constrained by the model's token limitations, particularly when handling large data queries for SVI statistics. This often led to the omission of context or truncated function responses. While prompt refinement to narrow the scope of queries offered a partial solution, future work should explore more robust engineering solutions, such as data chunking, pagination of results, or developing more efficient on-demand map tiling to circumvent these token limits.

Finally, two key areas require substantial future work before operational deployment. First, a detailed latency analysis (measuring P50/P90/P99 response times) is crucial to ensure the system can meet the demanding performance requirements of a real-time emergency. Second, while our user study validated usability, a dedicated user comprehension study is the logical next step. Such a study should measure improvements in map literacy, correct action selection, and overall risk perception compared to traditional methods. This would provide the empirical evidence needed to fully validate the "democratized engagement" claim and transition this promising prototype into a trusted, life-saving tool.

5. Conclusion and future work

This research addressed a critical gap between complex flood risk data and its effective communication by developing and validating a novel AI-powered assistant. Leveraging the capabilities of GPT-4 within a Retrieval-Augmented Generation framework, our work moves beyond traditional one-way data provision to create a dynamic, two-way conversational tool for risk interpretation. The key findings from our comprehensive evaluation quantitatively substantiate the promise and pinpoint the challenges of this approach. The assistant demonstrated high proficiency in handling general knowledge and alert-based queries, achieving near-perfect performance scores (5/5 and 4.3/5, respectively). However, its performance on more complex, data-intensive tasks like querying social vulnerability statistics was lower (4.0/5) and revealed significant latency issues, with response times for function calls averaging 36 s and peaking at 198 s for large dataset retrieval.

Our analysis of different models confirmed that GPT-4o offers the optimal balance of accuracy and speed, establishing a clear path for future implementation. The primary contribution of this work is a

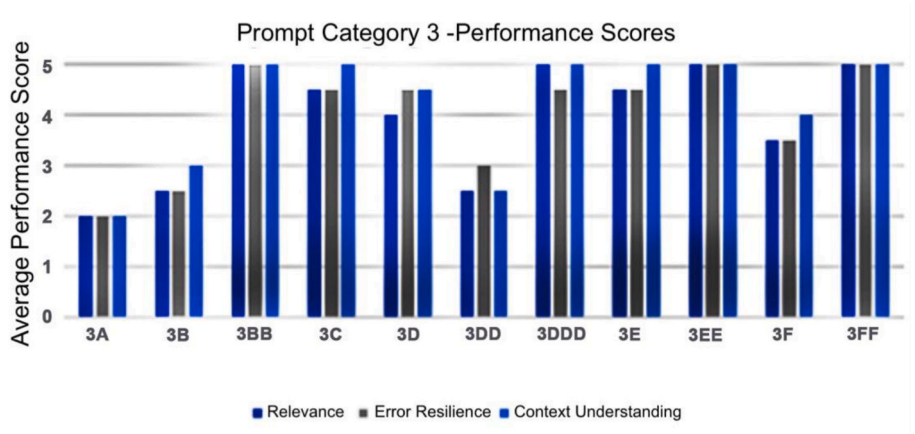


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validated methodology that proves LLMs can effectively translate technical data for non-experts, a conclusion supported by our user study, which showed significant improvements in task success and efficiency over a non-LLM baseline. While future work must focus on optimizing data-heavy functions and improving error resilience, this study provides a foundational blueprint for using LLMs to democratize environmental science. By transforming how risk information is communicated, this research paves the way for more resilient, informed, and equitably engaged communities in the face of escalating climate-related threats.

CRediT authorship contribution statement

Rafaela Martelo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kimia Ahmadiyehyazdi:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ruo-Qian Wang:** Writing – review & editing, Writing – original draft, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ruo-Qian Wang reports financial support was provided by Rutgers The State University of New Jersey. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envsoft.2025.106821.

Data availability

The data is shared in Github and the code is open source in Github.

References

Argent, R.M., Sojda, R.S., Giupponi, C., McIntosh, B., Voinov, A.A., Maier, H.R., 2016. Best practices for conceptual modelling in environmental planning and management. Environ. Model. Software 80, 113–121. https://doi.org/10.1016/j.envsoft.2016.02.023.

Arrighi, C., Pregnolato, M., Dawson, R.J., Castelli, F., 2019. Preparedness against mobility disruption by floods. Sci. Total Environ. 654, 1010–1022. https://doi.org/10.1016/j.scitotenv.2018.11.191.

Bodoque, J.M., Amérigo, M., Díez-Herrero, A., García, J.A., Cortés, B., Ballesteros-Cánovas, J.A., Olcina, J., 2016. Improvement of resilience of urban areas by integrating social perception in flash-flood risk management. J. Hydrol. 541, 665–676. https://doi.org/10.1016/j.jhydrol.2016.02.005.

Bodoque, J.M., Díez-Herrero, A., Amerigo, M., García, J.A., Olcina, J., 2019. Enhancing flash flood risk perception and awareness of mitigation actions through risk communication: a pre-post survey design. J. Hydrol. 568, 769–779. https://doi.org/10.1016/j.jhydrol.2018.11.007.

Botzen, W.J., Aerts, J.C., van den Bergh, J.C., 2009. Dependence of flood risk perceptions on socioeconomic and objective risk factors. Water Resour. Res. 45 (10). https://doi.org/10.1029/2009wr007743.

Bradford, R.A., O'Sullivan, J.J., van der Craats, I.M., Krywkow, J., Rotko, P., Aaltonen, J., Bonaiuto, M., De Dominicis, S., Waylen, K., Schelfaut, K., 2012. Risk perception – issues for flood management in Europe. Nat. Hazards Earth Syst. Sci. 12 (7), 2299–2309. https://doi.org/10.5194/nhess-12-2299-2012.

Bubeck, P., Botzen, W.J., Aerts, J.C., 2012. A review of risk perceptions and other factors that influence flood mitigation behavior. Risk Anal.: Int. J. 32 (9), 1481–1495. https://doi.org/10.1111/j.1539-6924.2011.01783.x.

Bullock, J., Luengo-Oroz, M., 2019. Automated speech generation from UN general assembly statements: mapping risks in AI generated texts. https://doi.org/10.48550/arXiv.1906.01946.

Chang, D.L., Yang, S.H., Hsieh, S.L., Wang, H.J., Yeh, K.C., 2020. Artificial intelligence methodologies applied to prompt pluvial flood estimation and prediction. Water 12 (12), 3552. https://doi.org/10.3390/w12123552.

David, A.O., Ndambuki, J.M., Muloiwa, M., Kupolati, W.K., Snyman, J., 2024. A review of the application of artificial intelligence in climate change-induced Flooding—Susceptibility and management techniques. CivilEng 5 (4), 1185–1198.

Forsyth, T., 2023. Climate change adaptation and resilience: sociotechnical and knowledge dimensions. In: Climate, Science and Society. Routledge, pp. 198–206.

Gillotte, J.L., 2019. Copyright infringement in AI-generated artworks. UC Davis Law Rev. 53, 2655. https://ssrn.com/abstract=365742.

Hagemeier-Klose, M., Wagner, K., 2009. Evaluation of flood hazard maps in print and web mapping services as information tools in flood risk communication. Nat. Hazards Earth Syst. Sci. 9 (2), 563–574. https://doi.org/10.5194/nhess-9-563-2009.

Henstra, D., Minano, A., Thistlethwaite, J., 2019. Communicating disaster risk? An evaluation of the availability and quality of flood maps. Nat. Hazards Earth Syst. Sci. 19 (1), 313–323. https://doi.org/10.5194/nhess-19-313-2019.

Hu, G., 2023. Challenges for enforcing editorial policies on AI-generated papers. Account. Res. 1–4. https://doi.org/10.1080/08989621.2023.2184262.

Hu, Y., Wang, R.Q., 2020. Understanding the removal of precise geotagging in tweets. Nat. Hum. Behav. 4 (12), 1219–1221. https://doi.org/10.1038/s41562-020-00949-x.

Köbis, N., Mossink, L.D., 2021. Artificial intelligence versus Maya Angelou: experimental evidence that people cannot differentiate AI-generated from human-written poetry. Comput. Hum. Behav. 114, 1–13. https://doi.org/10.1016/j.chb.2020.106553.

Lei, Z., Dong, Y., Li, W., Ding, R., Wang, Q., Li, J., 2025. Harnessing Large Language Models for Disaster Management: a Survey arXiv preprint arXiv:2501.06932.

Leskens, J.G., Brugnach, M., Hoekstra, A.Y., Schuurmans, W., 2014. Why are decisions in flood disaster management so poorly supported by information from flood models?

Environ. Model. Software 53, 53–61. https://doi.org/10.1016/j.envsoft.2013.11.003.
Li, L., Li, D., Ou, Z., Xu, X., Liu, J., Ma, Z., et al., 2025. In: LLMs as world models: Data-Driven and human-centered pre-event simulation for disaster impact assessment. Association for Computational Linguistics, Suzhou, China, pp. 3078–3096.
Liu, J., Wang, Y., Huang, X., Korsós, M.B., Jiang, Y., Wang, Y., Erdélyi, R., 2021. Reliability of AI-generated magnetograms from only EUV images. Nat. Astron. 5 (2), 108–110. https://doi.org/10.1038/s41550-021-01310-6.
Liu, Y., Han, T., Ma, S., Zhang, J., Yang, Y., Tian, J., He, H., Li, A., He, M., Liu, Z., Wu, Z., Zhao, L., Zhu, D., Li, X., Qiang, N., Shen, D., Liu, T., Ge, B., 2023. Summary of CHATGPT-related research and perspective towards the future of large language models. Meta-Radiol. 1 (2), 100017. https://doi.org/10.1016/j.metrad.2023.100017.
Maskrey, S.A., Mount, N.J., Thorne, C.R., Dryden, I., 2016. Participatory modelling for stakeholder involvement in the development of flood risk management intervention options. Environ. Model. Software 82, 275–294. https://doi.org/10.1016/j.envsoft.2016.04.027.
Meyer, V., Kuhlicke, C., Luther, J., Fuchs, S., Priest, S., Dorner, W., Serrhini, K., Pardoe, J., McCarthy, S., Seidel, J., Palka, G., Unnerstall, H., Viavattene, C., Scheuer, S., 2012. Recommendations for the user-specific enhancement of flood maps. Nat. Hazards Earth Syst. Sci. 12 (5), 1701–1716. https://doi.org/10.5194/nhess-12-1701-2012.
O'Sullivan, J.J., Bradford, R.A., Bonaiuto, M., De Dominicis, S., Rotko, P., Aaltonen, J., Waylen, K., Langan, S.J., 2012. Enhancing flood resilience through improved risk communications. Nat. Hazards Earth Syst. Sci. 12 (7), 2271–2282. https://doi.org/10.5194/nhess-12-2271-2012.
Pham, B.T., Luu, C., Phong, T.V., Nguyen, H.D., Le, H.V., Tran, T.Q., Ta, H.T., Prakash, I., 2021. Flood risk assessment using hybrid artificial intelligence models integrated with multi-criteria decision analysis in Quang Nam Province, Vietnam. J. Hydrol. 592, 125815. https://doi.org/10.1016/j.jhydrol.2020.125815.
Rollason, E., Bracken, L.J., Hardy, R.J., Large, A.R., 2018. Rethinking flood risk communication. Nat. Hazards 92 (3), 1665–1686. https://doi.org/10.1007/s11069-018-3273-4.
Strathie, A., Netto, G., Walker, G.H., Pender, G., 2017. How presentation format affects the interpretation of probabilistic flood risk information. J.Flood Risk Manag. 10 (1), 87–96. https://doi.org/10.1111/jfr3.12152.
Tapsell, S., McCarthy, S., Faulkner, H., Alexander, M., 2010. Social vulnerability to natural hazards. State of the Art Report from CapHaz-Net's WP4. London.
Wang, R.Q., 2021. Artificial intelligence for flood observation. Earth Observation Flood Appl. 295–304. https://doi.org/10.1016/b978-0-12-819412-6.00013-4.
Wang, R.Q., Ding, Y., 2023. Semi-supervised identification and mapping of surface water extent using street-level monitoring videos. Big Earth Data 7 (4), 986–1004.
Wang, R.Q., Mao, H., Wang, Y., Rae, C., Shaw, W., 2018. Hyper-resolution monitoring of urban flooding with social media and crowdsourcing data. Comput. Geosci. 111, 139–147. https://doi.org/10.1016/j.cageo.2017.11.008.
Whittaker, L., Kietzmann, T.C., Kietzmann, J., Dabirian, A., 2020. All around me are synthetic faces: the mad world of ai-generated media. IT Professional 22 (5), 90–99. https://doi.org/10.1109/mitp.2020.2985492.
Xie, Y., Jiang, B., Mallick, T., Bergerson, J.D., Hutchison, J.K., Verner, D.R., et al., 2025a. A rag-based multi-agent llm system for natural hazard resilience and adaptation. arXiv preprint arXiv:2504.17200.
Xie, Y., Jiang, B., Mallick, T., Bergerson, J., Hutchison, J.K., Verner, D.R., et al., 2025b. MARSHA: multi-agent RAG system for hazard adaptation. npj Climate Action 4 (1), 70.
Ye, X., Wang, S., Lu, Z., Song, Y., Yu, S., 2021. Towards an AI-driven framework for multi-scale urban flood resilience planning and design. Computational Urban Sci. 1 (1), 1–12. https://doi.org/10.1007/s43762-021-00011-0.
Zajac, M., Kulawiak, C., Li, S., Erickson, C., Hubbell, N., Gong, J., 2025. Unifying flood-risk communication: empowering community leaders through AI-Enhanced, contextualized storytelling. Hydrology 12 (8), 204.